

Report 13 Data Source Integration and Fallback Chains

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Report 13 — Data Source Integration: Registration, Wiring & Fallback Chains

Project: JalJeev — Agentic Marine Intelligence System
Phase: Ongoing (Aug-Sep 2026)
Focus: API registrations, adapter implementation, credential lifecycle, multi-source fallback

1. Overview

JalJeev integrates 15+ external data sources. This report documents the registration journey for each, the wiring pattern, and the fallback chains that ensure the system works even when any individual source is unavailable.

2. Data Source Registry & Status

2.1 Live and Verified

Source	Adapter	Auth	Status	First Data
Open-Meteo Marine	open_meteo_adapter.py	None	✔ Live	Wave, swell, wind, pressure
Open-Meteo Forecast	open_meteo_adapter.py	None	✔ Live	Weather forecast (wind, temp, precip)
NOAA OISST	noaa_oisst_adapter.py	None	✔ Live	Sea surface temperature
Copernicus Marine	copernicus_adapter.py	Username/password	✔ Live	Physics (SST, currents), BGC (chlorophyll)
ERA5/CDS	era5_adapter.py	API key	✔ Live	Historical atmospheric reanalysis
NASA Earthdata	CMR search	Credentials	✔ Live	Satellite granule discovery
GFW	gfw_adapter.py	API key	✔ Live	Historical fishing effort
MOSDAC (EOS-06)	mosdac_adapter.py	Username/password	✔ Active	OCM-3 chlorophyll
EOT20 (Tide)	tide_adapter.py	None (local model)	✔ Live	Astronomical tide prediction
GEBCO 2026	bathymetry_adapter.py	None (downloaded tiles)	✔ Live	Bathymetry (regional tiles)
Marine Regions	PostGIS (pre-loaded)	None	✔ Live	EEZ boundaries

2.2 Wired but Pending Activation

Source	Adapter	Status	Blocker
IMD	imd_adapter.py	☐ Pending API key	Registration submitted; no public API docs
Protected Planet	protectedplanet_adapter.py	☐ Pending API key	Research key requested
INCOIS PFZ Archive	—	☒ Blocked	Manual portal access required

2.3 Adapter Auto-Activation Pattern

Every adapter follows the same credential-check pattern:

```
async def get_imd_weather(lat, lon):
    if not settings.IMD_API_KEY:
        return {
            "status": "unavailable",
            "source": "imd",
            "reason": "IMD_API_KEY not configured"
        }
    # ... actual API call
```

Dropping a key into `.env` activates the adapter with zero other code changes. The calling agents already have the fallback chain built.

3. Fallback Chains (Per Variable)

3.1 Wave Height

INCOIS OSF (ERDDAP) → Copernicus Marine (physics) → Open-Meteo Marine

Currently: INCOIS SSL-broken, Copernicus slow, **Open-Meteo is the workhorse**.

3.2 SST (Live)

INCOIS → Copernicus Marine → NOAA OISST → MOSDAC (EOS-06)

Currently: NOAA OISST + Copernicus Marine operational.

3.3 Wind / Weather

IMD (primary, pending) → Open-Meteo Forecast

Currently: Open-Meteo Forecast is the primary and only source.

3.4 Chlorophyll

MOSDAC (EOS-06 OCM-3) → Copernicus Marine (BGC) → NASA OceanColor

Currently: Copernicus Marine BGC operational; MOSDAC activated.

3.5 Cyclone / Severe Weather

IMD (primary, pending) → IBTrACS (historical) → none

Currently: **No live cyclone data**. This is a known gap.

3.6 Geofencing

PostGIS (Marine Regions EEZ) – always available
PostGIS (Protected Planet MPA) – pending API key activation

3.7 Tide

EOT20 (local model) – always available

No external dependency — the harmonic model runs locally.

3.8 Bathymetry

GEBCO tiles (local) – available for loaded regions only

Points outside loaded tiles return "unavailable", never zero.

4. Circuit Breaker (`_circuit_breaker.py`)

4.1 Purpose

If a data source fails repeatedly, the circuit breaker stops trying for a cooldown period to avoid: - Wasting latency on known-broken sources - Flooding external APIs with failing requests - Degrading overall response

time

4.2 Implementation

```
class CircuitBreaker:
    FAILURE_THRESHOLD = 5      # Open circuit after 5 consecutive failures
    RECOVERY_TIMEOUT = 300    # Try again after 5 minutes

    def is_open(self, source_name: str) -> bool:
        """Return True if we should skip this source."""

    def record_failure(self, source_name: str):
        """Track a failure."""

    def record_success(self, source_name: str):
        """Reset the failure counter."""
```

4.3 Integration

Every adapter checks the circuit breaker before making an external call:

```
if circuit_breaker.is_open("copernicus_marine"):
    return {"status": "failed", "source": "copernicus_marine", "reason": "circuit breaker open"}
```

5. Redis Cache Architecture (`core/cache.py`)

5.1 Per-Variable TTLs

Variable	TTL	Rationale
SST	6 hours	Changes slowly
Wave forecast	3 hours	Forecast updates ~6-hourly
Wind forecast	3 hours	Same
Tide series	12 hours	Astronomical, deterministic
Cyclone track	30 minutes	Safety-critical, fast-changing
EEZ/MPA containment	24 hours	Static boundaries
Bathymetry	∞ (startup load)	Never changes

5.2 Cache Key Design

```
cache_key = f"jaljeev:{source}:{variable}:{h3_cell_res7}:{time_bucket}"
```

Using H3 cells in cache keys means nearby queries hit the same cache entry — a query at 17.649°N, 83.351°E hits the same cache as one at 17.651°N, 83.349°E (same H3 cell).

6. The ERDDAP Broken Import (R-11)

A version mismatch in the virtual environment broke `erddapy` :

```
cannot import name '_quote_string_constraints'
```

This also breaks the argo/ERDDAP xarray backends at import. NOAA OISST currently succeeds from cache but

its live path is suspect.

Fix needed: Version pin in requirements.txt .

7. Registration Timeline

Source	Registered	Activated	Days
Copernicus Marine	Aug 25	Aug 25	Same day
ERA5/CDS	Aug 25	Aug 25	Same day
NASA Earthdata	Aug 25	Aug 25	Same day
GFW	Aug 25	Aug 29	4 days
MOSDAC	Aug 25	Aug 27	2 days
IMD	Aug 25	☐ Pending	—
Protected Planet	Aug 26	☐ Pending	—
Groq	Aug 26	Aug 26	Same day

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